

NexaQuant

Evidence-First Quant Trading System

Gold (XAUUSD) + Bitcoin (BTCUSD) · Smart Money Concepts + Regime Gate + AI

TECHNICALS + SMC

structure · FVG · order blocks
liquidity · premium/discount

REGIME + RISK

trend/range/volatile gate
ATR stops · vol-target sizing

AI + FUNDAMENTALS

meta-labeling · MTF features
macro bias · ensemble-ready

Architecture & Evidence Deck

Charts in this deck are generated live from the real 2y gold dataset and the project's own backtests.

Prove the edge, control drawdown, gate capital behind validation

The Thesis

Profit = edge x payoff x risk control — NOT win-rate or model hype. We capture a small, real, cost-robust edge (trend-continuation incl. SMC structure/FVG), cut drawdown hard with a regime gate, and let AI sharpen selection/sizing as data & features grow. Capital scales only behind a hard validation gate.

Regime-gated trend (H1)

Sharpe 3.6

out-of-sample, net of cost

Drawdown vs buy&hold

-33%

254 – > 172 (H1 OOS)

FVG + Structure (H4)

Sharpe 1.9

robust IN & OUT of sample

AI meta-label today

AUC 0.51

premature — needs data+features

VALIDATED (out-of-sample, net of cost)

+ Trend-continuation edge (EMA + FVG + Break of Structure)

+ Regime gate lifts Sharpe & cuts drawdown ~33%

+ Edge survives realistic spread/slippage

+ Multi-timeframe + fundamental feature pipeline wired & verified

NOT YET PROVEN (honest gaps)

- Much of return is GOLD BETA (2023-25 bull) — one regime only

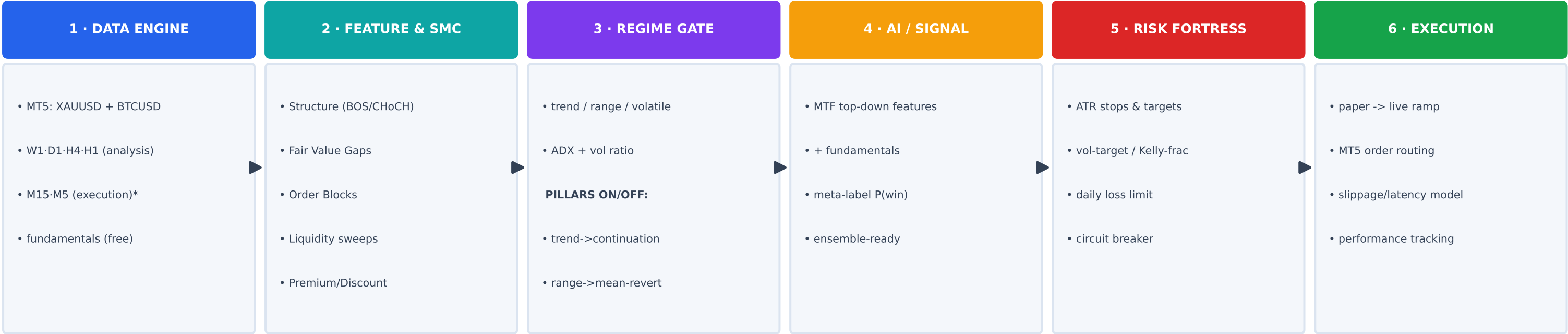
- Mean-reversion SMC (discount, sweep-fade) lost in trend

- AI flat until M5/M15 + BTC + fundamentals add data/features

- No live/paper validation yet — do NOT deploy capital

The bet: invest in DATA + FEATURES + the VALIDATION GATE — that, not one magic model, is the asset.

Six layers — nothing reaches live capital without passing the gate

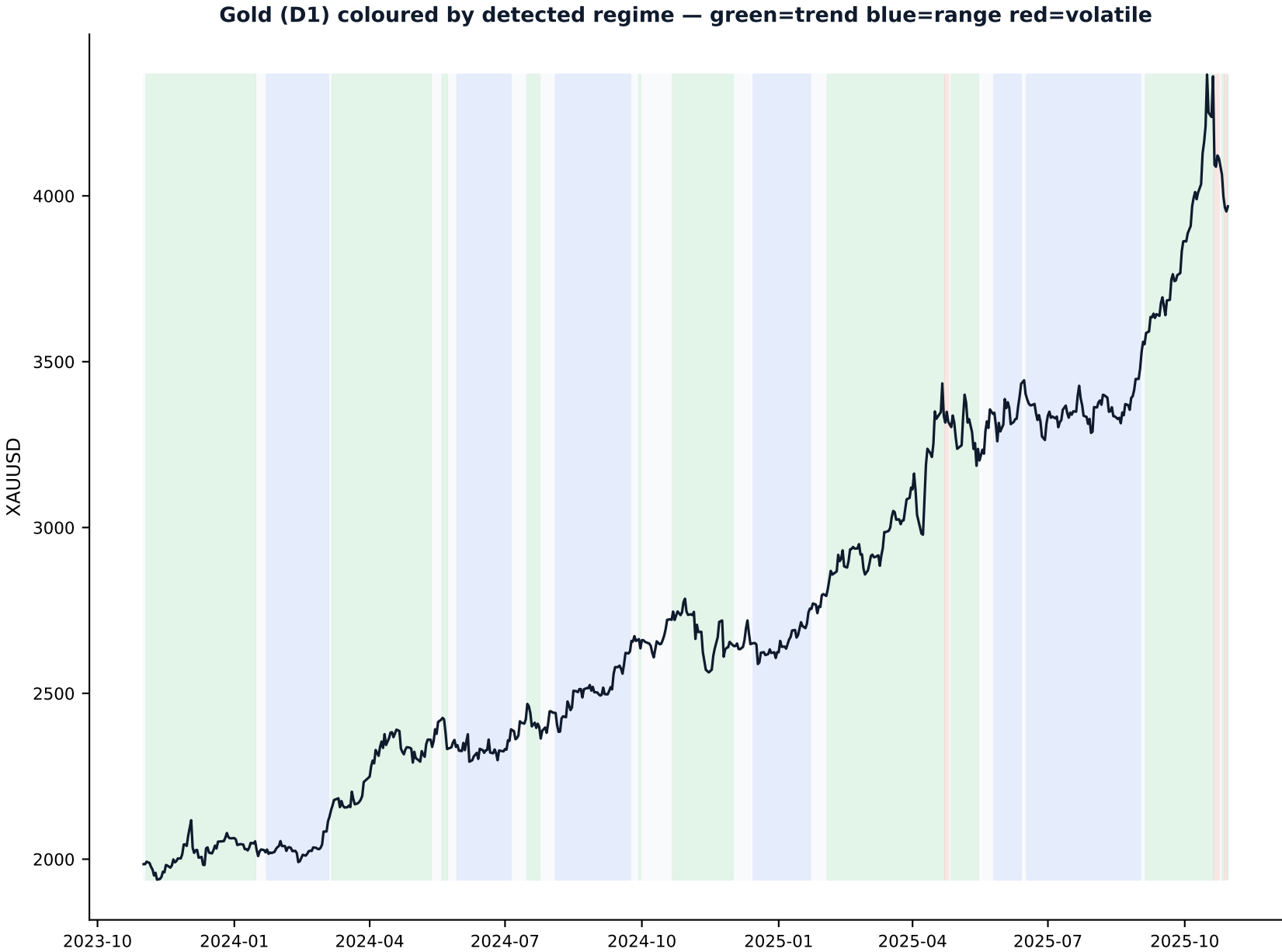
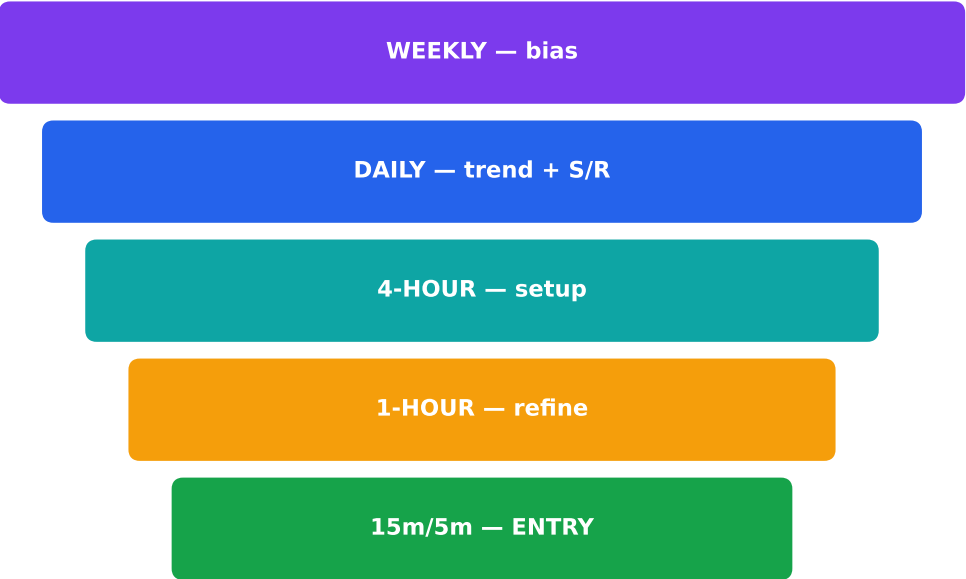


Analyse high, execute low — higher TFs become model features too

Instruments & timeframes

symbol	analysis	execution
XAUUSD (gold)	W1 D1 H4 H1	M15 M5
BTCUSD	W1 D1 H4 H1	M15 M5
Fundamentals (FREE): FRED real yields · DXY · CFTC COT · economic calendar		

Top-down funnel (also the model's MTF features)



Read institutional footprints — the regime decides which to trust

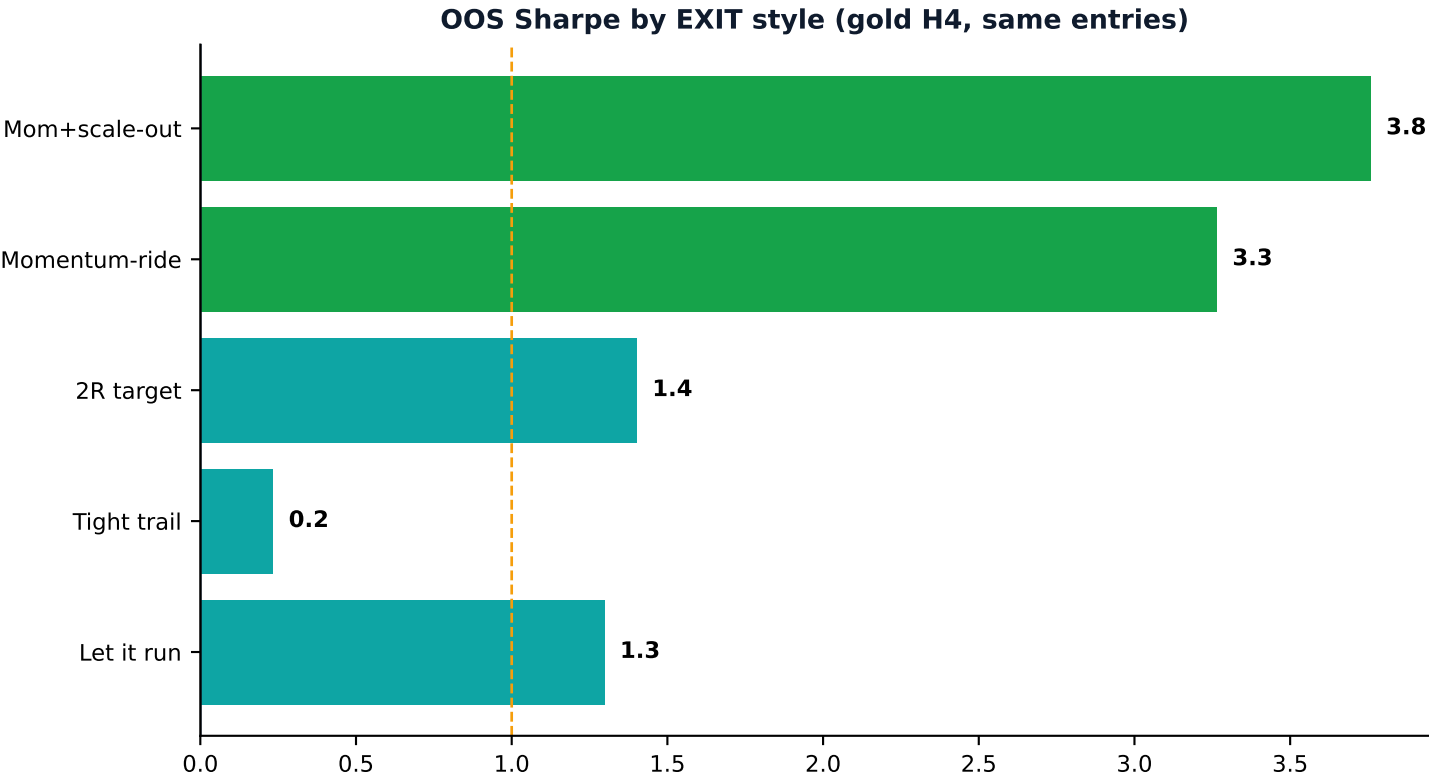
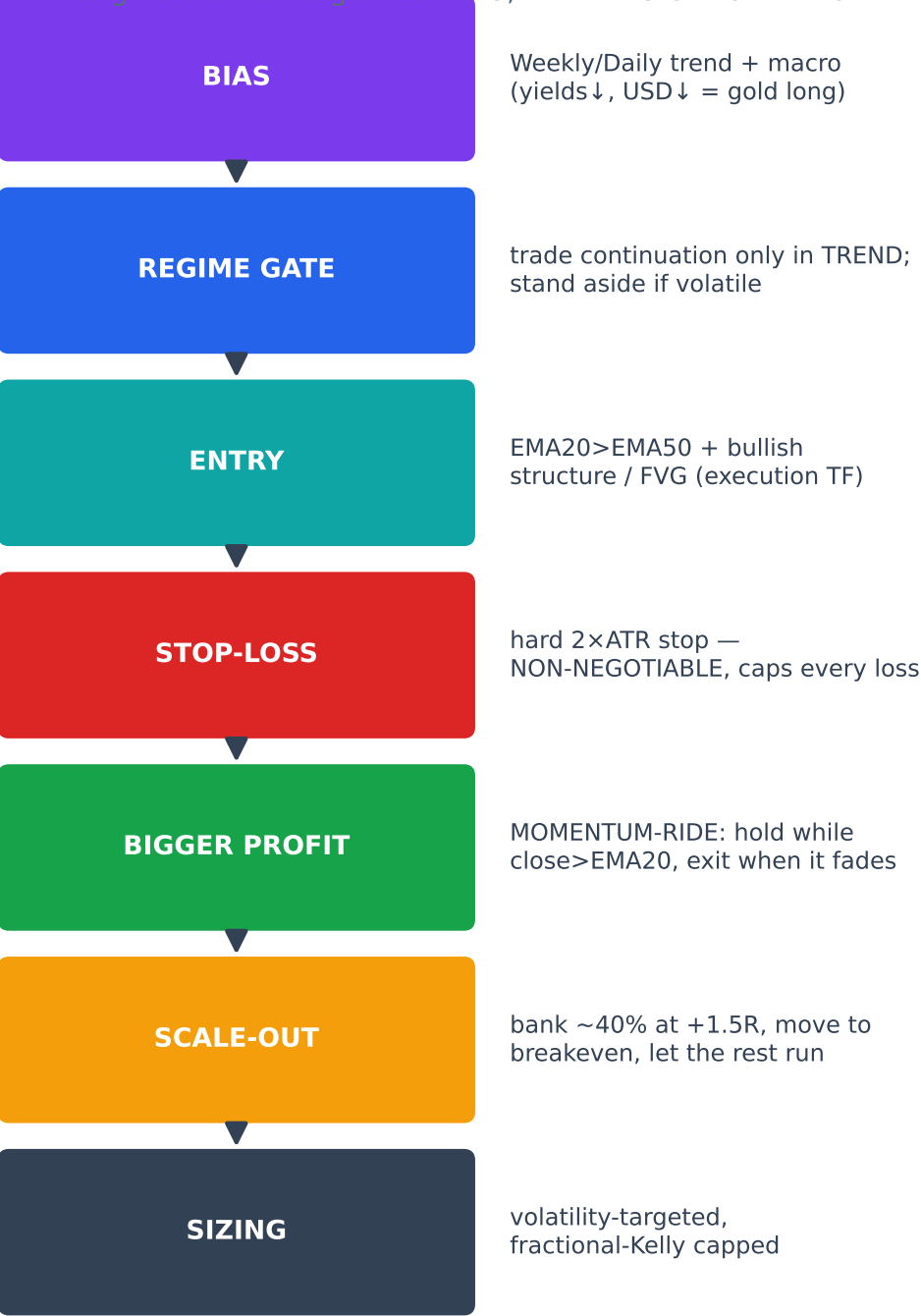
The 5 pillars (with measured edge)		
1 • Fair Value Gap	3-candle imbalance; price revisits	EDGE +
2 • Market Structure	BOS / CHoCH = trend & reversal	EDGE +
3 • Order Blocks	last opposite candle = demand/supply	context
4 • Liquidity Sweeps	stop-hunt beyond swing then reclaim	EDGE - in trend
5 • Premium / Discount	25/50/75% of swing range	EDGE - in trend

Regime gate decides
TRENDING
ON: Structure + FVG (continuation) — GO WITH the move
RANGING
ON: Discount-buy + sweep (mean-revert) — FADE extremes
VOLATILE
size DOWN or STAND ASIDE

KEY INSIGHT FROM OUR DATA

Continuation pillars (Structure + FVG) had POSITIVE edge on trending gold; mean-reversion pillars (deep-discount, sweep-fade) had NEGATIVE edge. Stacking all 5 ('A+') LOST money — the regime gate, not more indicators, is what turns SMC into profit.

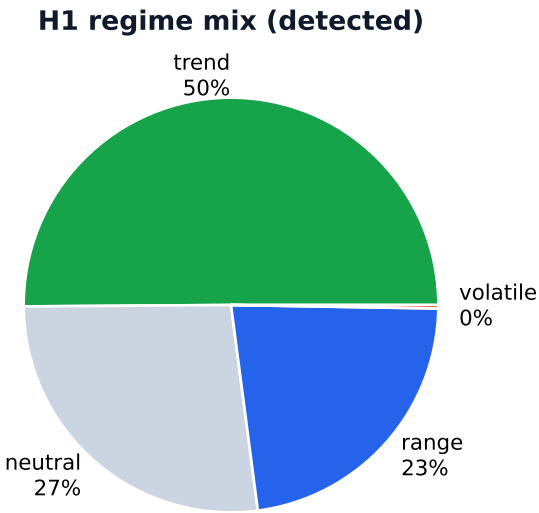
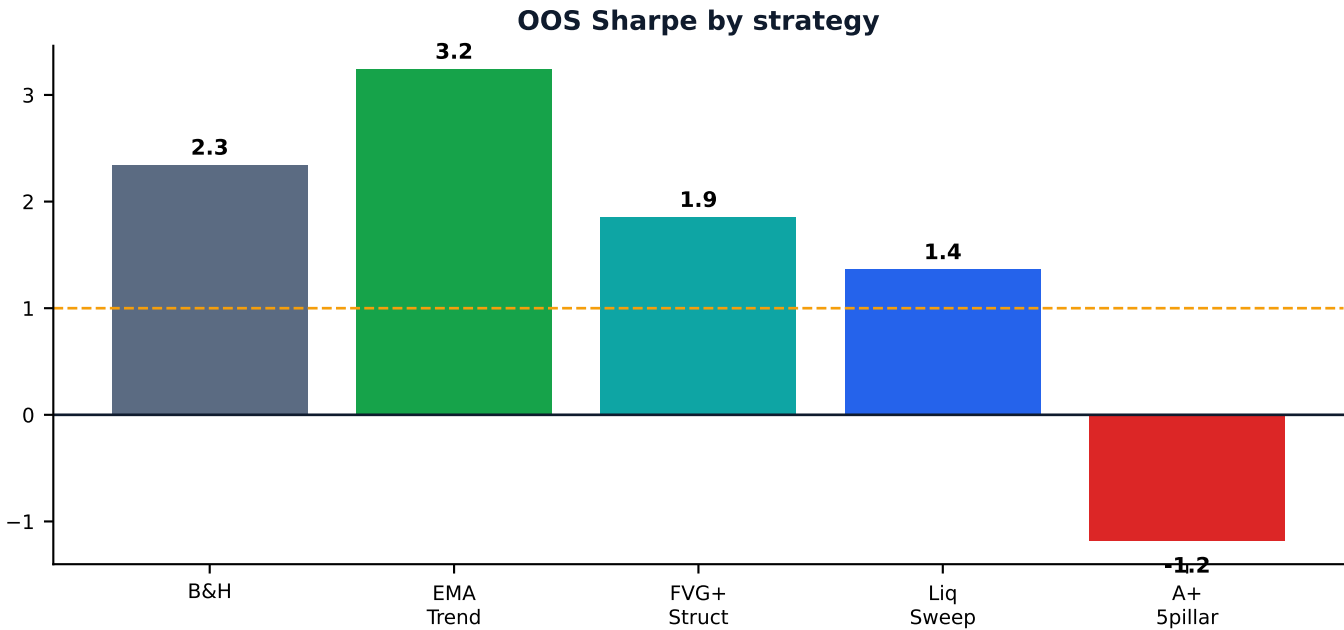
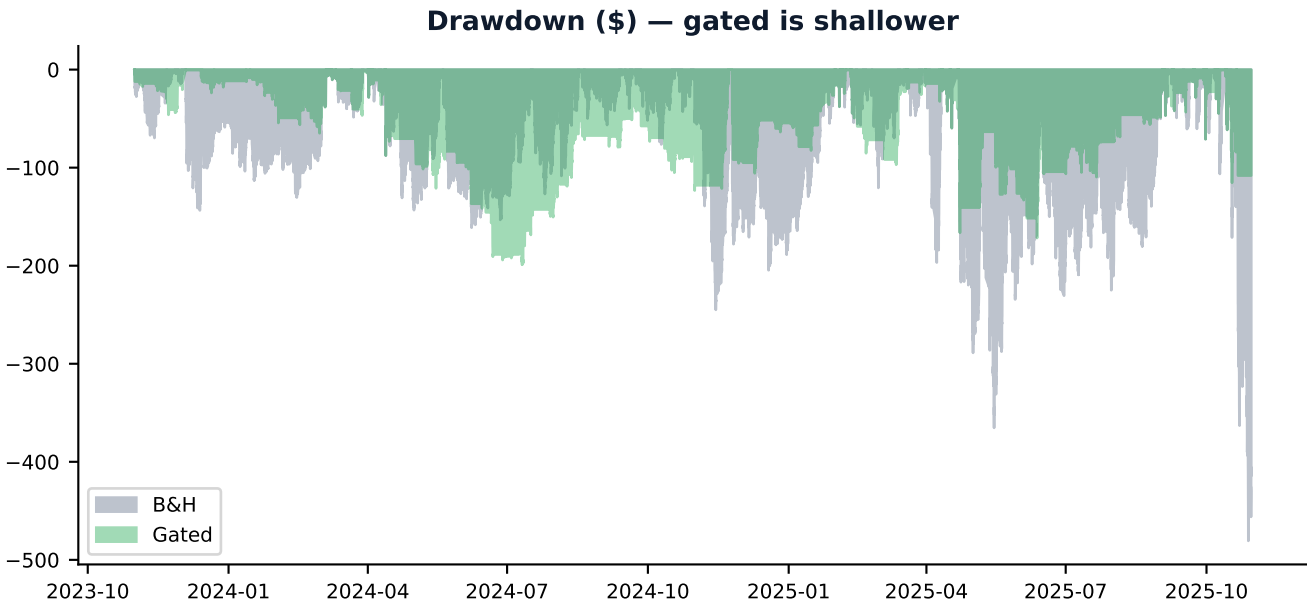
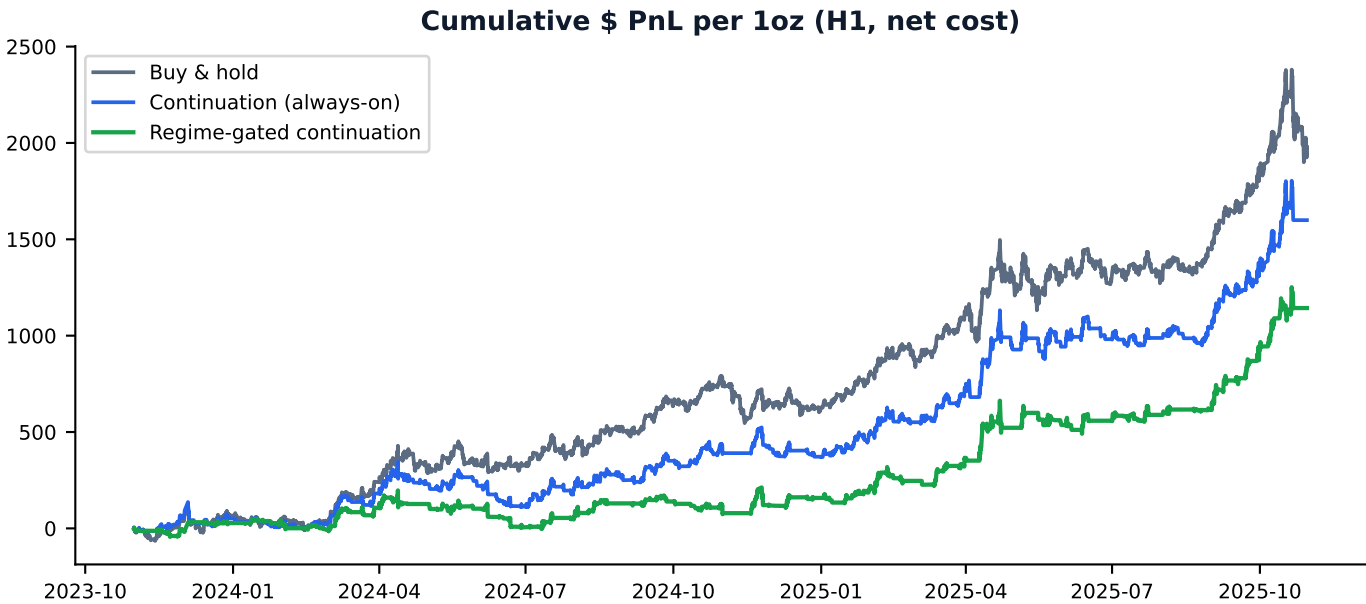
Entries give a small edge — STOPS, TRAILING & MOMENTUM-RIDING make the profit



WHY MOMENTUM-RIDE

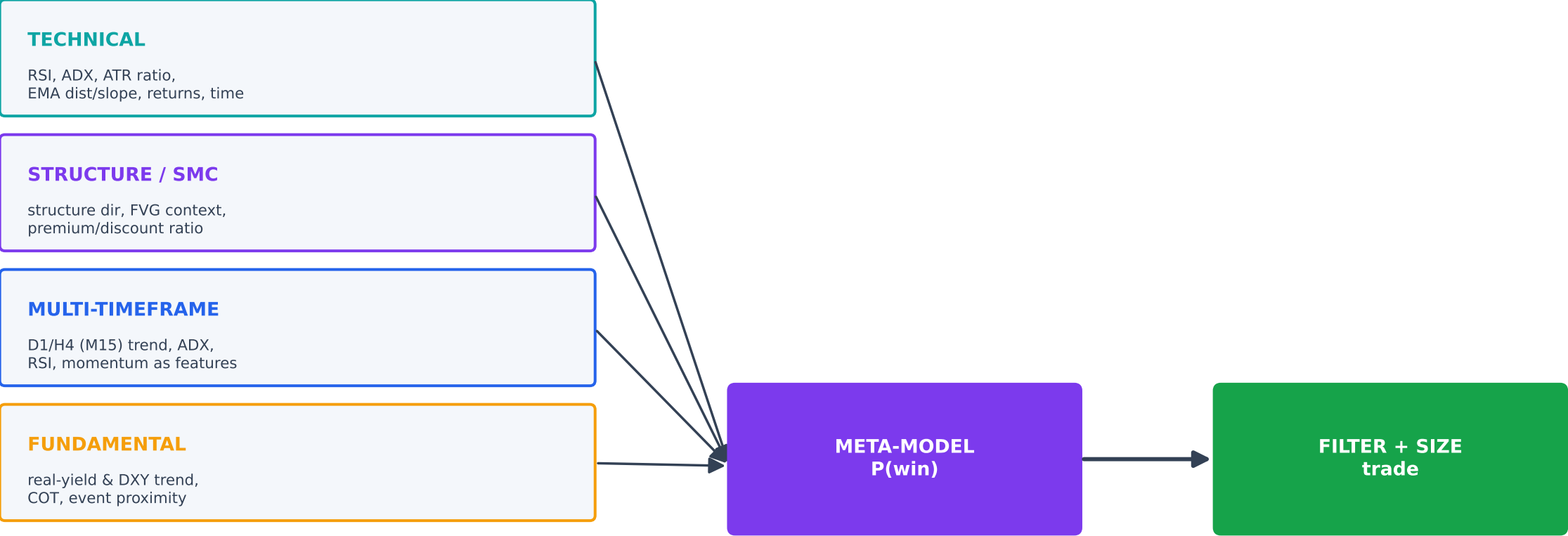
- Stop-loss is mandatory — caps every loser
- Tight trailing turns the edge NEGATIVE (chokes winners)
- Riding momentum captures big R-multiples (up to 8.7R) while cutting drawdown ~40% vs 'let it run forever'
- Scale-out lifts win-rate to ~64% and smooths the curve

Computed live from the real 2y gold data + project backtests



Models SIZE & SELECT a known edge — they do not invent one

Feature pipeline -> meta-model -> sizing



Today: AUC 0.51 (coin-flip) — 154 entries, technical-only. The plumbing is verified (28/28 features active with MTF+fundamentals wired); it gains skill as data/features grow.

Model staging (complexity rises only with data + validation)



Why most backtests fail live — and how we refuse to

The 5 ways backtests lie (and our defence)

Data leakage / look-ahead

signal at close(t) -> trade at open(t+1); HTF & macro shifted to release

Overfitting / curve-fit

default params; in-sample vs out-of-sample split always reported

Multiple testing

few strategies tested; deflated-Sharpe mindset; no p-hacking

Ignoring costs

every trade net of spread+slippage; cost-sensitivity run

One-regime sample

regime-stratified results; flagged as a known gap

The gate (in order)

Temporal firewall (embargo)



Walk-forward + Purged CV



Regime-stratified backtest



30-day paper trading



Calibration (Brier<0.25)



Go / No-Go certification

BUILDING NEXT: walk-forward + Combinatorial Purged CV (López de Prado) — the rigor gate that lets us trust ANY strategy or model before capital.

What the peer-reviewed literature actually supports (deep-research, adversarially verified)

Topic	Finding	Evidence	NexaQuant decision
Stop-loss	Regime-conditional: HELPS under momentum, HURTS under random-walk/mean-reversion <i>Kaminski & Lo 2014</i>	STRONG	VALIDATES us — we only stop within the TREND regime
Let winners run	Fixed/MACD take-profit did NOT beat letting it run (FX/metals/crypto) <i>Vezeris 2018</i>	MODERATE	VALIDATES — momentum-ride > 2R target in our tests
Stop+momentum	10% stop on momentum: Sharpe 0.17->0.37, skew flipped +; tail capped <i>Han-Zhou-Zhu 2016</i>	STRONG*	Supports SL + momentum (but equity, gross of cost)
Vol-targeting	NAIVE single-factor vol-targeting fails OOS & net of costs <i>Cederburg 2020</i>	STRONG(-)	CAUTION — matches our cost-churn finding; size cost-aware
Vol (multifactor)	Multifactor, cost-optimized vol-conditioning DOES add ~13% Sharpe <i>DeMiguel 2024</i>	STRONG	Adopt at portfolio level, not naive per-bar
CPCV + Deflated SR	Adaptive search -> spurious backtests; needs trial-adjusted metrics <i>Bailey & LdP; Harvey-Liu</i>	STRONG	VALIDATES our validator.py (CPCV + DSR + embargo)
Deep RL	Crypto DRL prone to false-positive overfit; needs overfit test <i>Gort et al. 2022</i>	MODERATE	VALIDATES deferring RL behind the gate
SMC (FVG/OB/ICT)	ZERO verified academic evidence of independent edge <i>(unanswered)</i>	NONE	Treat as folklore; our edge = trend/momentum overlap
Free data	Dukascopy, CFTC COT, FRED are primary/trusted sources <i>Dukascopy/CFTC/FRED</i>	n/a	Our pullers target these (+ Binance, Stooq)

SCOPE CAVEAT + WHAT TO AVOID

Most STRONG evidence is monthly US EQUITY factors — NOT intraday gold/BTC. Mechanisms transfer, but we must RE-VALIDATE on our instruments (the rigor gate does this).
Refuted / avoid: specific ATR(12,6,2) params · Kelly-VIX hybrid sizing · 'vol-scaling doubles t-stat' · CVaR-variant superiority claims.

Design goal: make money at LOW RISK by trading only high-confidence setups

DESIGN PRINCIPLE

Win-rate / trade-selection first; lot size follows. Regime-aware LONG+SHORT, confluence entries (trend regime + structure + momentum), momentum-ride exit + scale-out, CONFIDENCE-TIERED risk (0.5% / 1% / 2% by trend strength, 2% hard cap). AI meta-label tightens selection as data grows.

Re-test — best pair & timeframe (OOS, net of cost)

pair	TF	win%	Sharpe	maxDD	verdict
BTC	H1	50%	3.66	2.5%	low-risk pick
XAU	H1	44%	4.59	3.9%	low-risk pick
BTC	H4	43%	1.95	14.8%	more return, more DD
ETH	H4	46%	-0.01	28%	weak
BNB	H1	26%	0.40	5.3%	weak
SOL	H4	42%	-2.41	33%	avoid

REALISTIC TARGET

Low risk, high Sharpe: BTC/XAU H1, long+short.
~3% max drawdown, win ~50% with big payoff.
Return ~10-25%/yr (more with capital + AI selection).
Beats a bank 3-5x at low drawdown.

HONEST LIMITS

5-10%/MONTH is NOT achievable sustainably (Medallion, the best ever, ~4%/mo). Chasing it = ruin. Bigger money = robust edge x CAPITAL x time, NOT leverage. Backtest, not live — paper-test first.

Honest, evidence-first: every idea back-tested on BTC H4, kept only if it provably helps

KEPT — proven to add pips

LENGTHY-CANDLE CONFIDENCE BOOST

A wide-range bar (body >= 1.5x ATR) confirming a regime entry marks the highest-quality trades (52% win, PF 3.38, 1,444 avg pips vs 84 baseline). Used as a +50% SIZE boost (not a standalone trigger): BTC H4 return +42% -> +54% at the SAME 4% drawdown. Big candles confirm; they don't trigger.

REJECTED — tested, did not survive (won't be shipped)

Pyramiding (add-to-winner)

+42% -> -13%, drawdown 4% -> 20% — adds buy near tops

Candle streaks 2/3/4 as entry

profit factor < 1.0 on EVERY timeframe — patterns fire too late

Pure expansion candle entry

PF 0.96 — lengthy candle works only as a filter, not a trigger

Fast timeframes (M5 / M15)

M5 = -100% (account wiped), M15 = -77% — cost + noise dominate

WEEKLY SELF-LEARNING LOOP (execution/auto_update.py)

Runs every week (systemd timer): 1) PULL fresh bars (incremental, history grows) 2) RE-VALIDATE via anchored PER-YEAR walk-forward (not a single noisy slice) 3) PROMOTE a config only if it passes the gate AND beats the champion — never auto-degrades 4) write the bot's TRADING LICENSE (health.json). The live bot reads the license each cycle: if the edge stops persisting on fresh data, it AUTO-STANDS-DOWN to manage-only. Self-learning with a guardrail — it cannot keep trading a dead strategy.

Research-sourced ideas, each tested under the rigor gate — kept only if it provably helped

ADOPTED (gold-only — BTC entries already clean)

Multi-lookback TSM confirmation require 20/60/120/240-bar momentum agreement	gold Sharpe -0.14 -> +0.55
Fundamental macro gate real yields + DXY falling = gold-bullish (free data)	stacks: Sharpe 0.19 -> 0.68, win 38->44%

REJECTED (tested, did not beat the champion)

Volatility-targeting overlay already implicit in our ATR-stop risk sizing	
Crash-protection de-risk over-trims winners; tiny DD gain, lower Sharpe	
Meta-label as AI sizer CPCV AUC ~0.50 = no skill yet -> keep ADX confidence	
Mean-reversion sleeve (ranges) loses; ranges break into trends on BTC/gold	

LAST 3 YEARS (0.5%-base Option B, net of cost)

BTC H4: 3/3 years profitable, +8.5% compounded, drawdown <= 4.3%. Gold H4: thin history (~2y), +5.1%. Lever for bigger returns = base risk (tiers already scale to 2% on strong trades; ~linear).

HONEST VERDICT

The bottleneck is DATA, not techniques: sizing overlays don't help an already risk-normalised book; signal-quality filters (TSM, macro) help GOLD. AI meta-label has no skill yet (needs deeper history). Edge is real but regime-dependent — paper-trade first.

Diversification across UNCORRELATED edges = the only free lunch; the path to consistent long-term

WHY A PORTFOLIO

One edge is regime-dependent — when trends die it idles. Combining several edges that are NOT correlated means a drawdown in one is cushioned by the others. Equal-risk (inverse-vol) weighting, config-driven (edges block), each sleeve validated independently and watched for decay by the weekly loop.

Single edge vs 4-sleeve portfolio (per-year, net of cost, Option B)

metric	trend:BTC only	PORTFOLIO (4 sleeves)
Full 5y return	+24.3%	+34.2%
Last 3y compounded	+8.5%	+10.8%
Max drawdown	4.3%	5.3%
Profitable years	6/6	6/6
Return sources	1 (fragile)	4 uncorrelated (durable)

VALIDATED EDGES (in the book)

1. Time-series TREND (EMA momentum-ride) — BTC, gold

2. Volatility BREAKOUT (Donchian) — BTC, gold

Cross-sleeve correlation ~0.0-0.3 = genuinely independent. Gold also gated by TSM + macro.

REJECTED / NEXT

Cross-sectional momentum: 82% DD (needs a BROAD universe, not 4 correlated cryptos).
Path to 10/10: more instruments + edges (more data), CCXT crypto infra, and a LIVE track record.

Confidence-scaled lots, earned by account growth — never hardcoded, never ruinous

OPTION B — how every lot is sized (execution/live_trader_calc_lots)

lot = (live_balance x risk% x CONFIDENCE) / broker_loss_per_lot [read live from MT5 symbol_info]
risk% tiers 0.5/1/2% by trend strength; CONFIDENCE = ADX x lengthy-candle boost. Re-reads balance each cycle: compounds up as profit grows, shrinks on withdrawal/drawdown. Zero hardcoding; any broker/account.

\$10 account, 3-yr BTC H4 — fixed lots vs Option B

fixed 0.01 (standard acct)	RUINED on trade #1 — one stop > the whole \$10
fixed 0.05 (cent acct)	+49% but 46% drawdown — pure leverage, no skill
Option B %-risk compounding	+9% at 4% drawdown — best risk-adjusted, can't be ruined

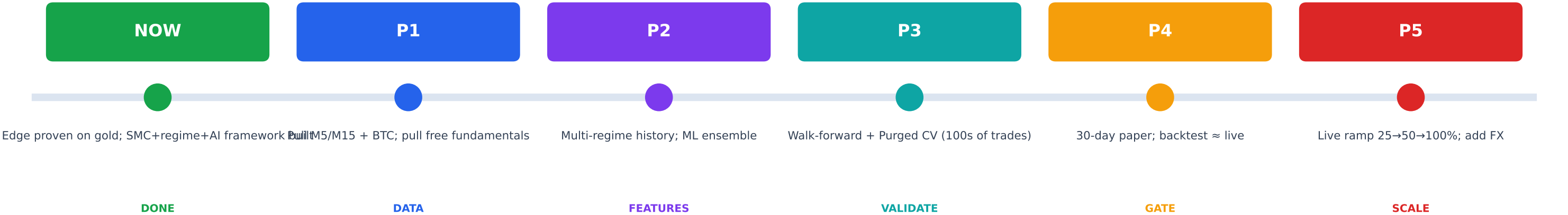
WHEN bigger lots (0.02/0.04/0.05) apply: as the balance GROWS (~\$20/\$40/\$50 -> next tier) AND on HIGH-CONFIDENCE trades (strong ADX + lengthy candle multiply the lot 2-4x). Earned, not forced onto \$10.

SELF-LEARNING + SAFETY (execution/auto_update.py -> health.json -> the bot)

Weekly (systemd timer): pull fresh bars -> re-validate via anchored PER-YEAR walk-forward -> promote a config only if it BEATS the champion (never auto-degrades) -> write the bot's TRADING LICENSE.
The live bot reads the license each cycle and AUTO-STANDS-DOWN to manage-only if the edge stops persisting on fresh data or the check goes stale (>21d). Risk guards: ATR stop (non-negotiable), daily-loss + drawdown kill switch, vol-spike/news blackout, tiny-account feasibility skip.
Self-learning WITH a guardrail — it can never keep trading a dead strategy or over-risk a small account.

The only safe path to live capital

Phases



Live targets (realistic, post-cost)

50-60%	1.0-1.5	1-4%	< 20%	90-95%
Win rate	Sharpe	Monthly ROI	Max drawdown	Uptime

SUCCESS METRIC

A strategy that holds Sharpe $\geq$ 1.0 ACROSS regimes AND in 30-day paper trading earns real capital — nothing else does. Model complexity (ML $\rightarrow$ ensemble $\rightarrow$ RL $\rightarrow$ multi-agent) increases ONLY together with data breadth and validation rigor.